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OVERVIEW OF HARDWARE COMPONENTS THEORY

Introduction to motherboards, CPUs, RAM,
GPUs, storage (HDD, SSD).

Importance
of

Each component in a computer system.

Assignment:

Create a diagram of a PC with
labeled hardware components

TASKS :- 1

Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

PARTS OF A COMPUTER

PSU



RAM



HDD



CPU COOLER



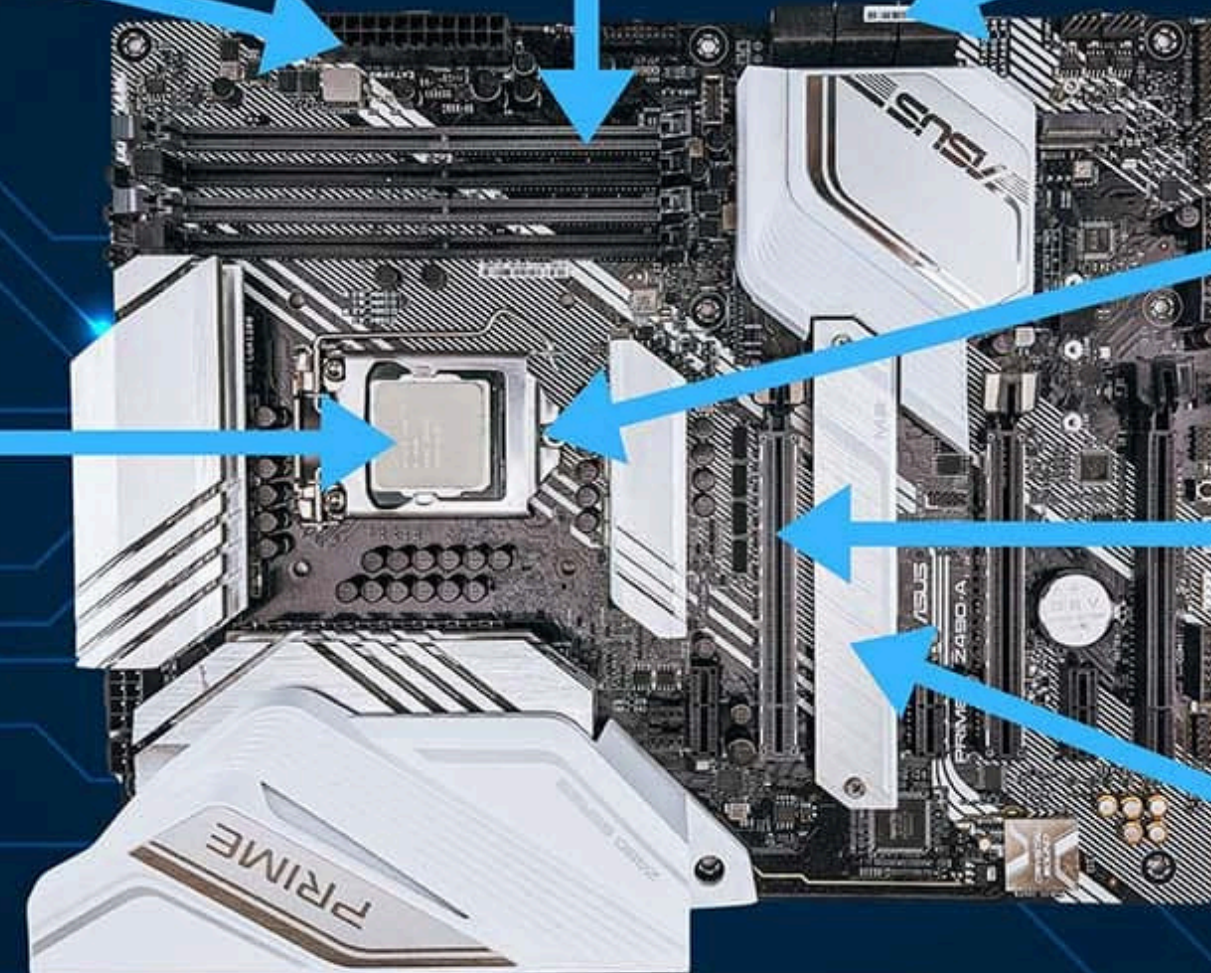
CPU



GRAPHICS CARD

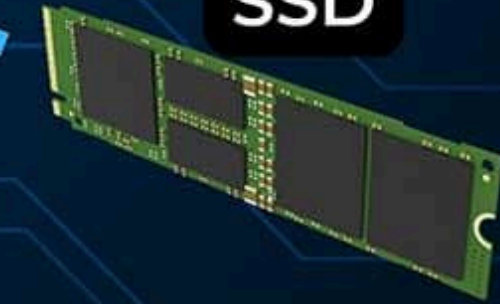


COMPUTER CASE



MOTHERBOARD

SSD



COOLING FANS (INSIDE CASE)



TASKS :-2

Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases

	FEATURES	HDD	SSD
1.	speed	Slower; uses spinning disks.and mechanical read/write head	Much faster; uses flash memory with no moving parts
2.	Durability	Less durable; mechanical parts can be damaged by shocks or drops	More durable; better Resistant to physical shocks
3.	Typical use cases	Large-capacity,low-cost storage Such as movies,photos,backups and archives	Operating system,application gaming,laptops,and tasks needing fast performance
4.	Noise	can make noise	silent
5.	Laptop	common in older laptop	common in modern laptop
6.	Gaming pc	good for storing large game libraries	best for faster game loading

DIFFERENCE B/W HDD AND SSD

TASKS:- 3

describing the role of the CPU,
RAM, and GPU in running a
game like PUBG or Free Fire
on a PC.

CPU

The CPU (Central Processing Unit) handles the main instructions and calculations of the game. It manages game logic, player movements, physics, and communicates with other hardware.

RAM

RAM (Random Access Memory) temporarily stores the game's data and instructions while the game is running. More RAM helps the game run smoothly and reduces slowdowns when other programs are open.

GPU

The GPU (Graphics Processing Unit) renders the game's graphics, such as characters, maps, textures, and effects. A powerful GPU provides higher frame rates and smoother visuals, especially at higher graphics settings.

TASKS:-4

Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each

Hardware components	Why upgrade it
1. GPU(GRAPHIC CARD)	A powerful GPU helps render FIFA's graphics smoothly and provides higher FPS at better graphics settings.
2. cpu(processor)	A faster CPU improves game calculations, player movements, and overall game performance
3. RAM	Upgrading to 16 GB or more helps the game run smoothly, especially when other applications are running in the background.
4. SSD	An SSD provides faster game loading and reduces loading times compared with an HDD.
5. Cooling system	Better cooling helps prevent the CPU and GPU from overheating during long gaming sessions.